

# **Technical Description**

**ENERCON SCADA PDI-OPC**

**ENERCON EP1, EP2, EP3, EP4 Wind Energy Converters**

**Publisher**

ENERCON GmbH ▪ Dreekamp 5 ▪ 26605 Aurich ▪ Germany  
Phone: +49 4941 927-0 ▪ Fax: +49 4941 927-109  
E-mail: [info@enercon.de](mailto:info@enercon.de) ▪ Internet: <http://www.enercon.de>  
Managing Directors: Hans-Dieter Kettwig  
Local court: Aurich ▪ Company registration number: HRB 411  
VAT ID no.: DE 181 977 360

**Copyright notice**

The entire content of this document is protected by copyright and – with regard to other intellectual property rights – international laws and treaties. ENERCON GmbH holds the rights in the content of this document unless another rights holder is expressly identified or obviously recognisable.

ENERCON GmbH grants the user the right to make copies and duplicates of this document for informational purposes for its own intra-corporate use; making this document available does not grant the user any further right of use. Any other duplication, modification, dissemination, publication, circulation, surrender to third parties and/or utilisation of the contents of this document – also in part – shall require the express prior written consent of ENERCON GmbH unless any of the above is permitted by mandatory legislation.

The user is prohibited from registering any industrial property rights in the know-how reproduced in this document, or for parts thereof.

If and to the extent that ENERCON GmbH does not hold the rights in the content of this document, the user shall adhere to the relevant rights holder's terms of use.

**Registered trademarks**

Any trademarks mentioned in this document are intellectual property of the respective registered trademark holders; the stipulations of the applicable trademark law are valid without restriction.

**Reservation of right of modification**

ENERCON GmbH reserves the right to change, improve and expand this document and the subject matter described herein at any time without prior notice, unless contractual agreements or legal requirements provide otherwise.

**Document details**

|                    |                                                                               |            |                                                        |
|--------------------|-------------------------------------------------------------------------------|------------|--------------------------------------------------------|
| <b>Document ID</b> | D0141306-9                                                                    |            |                                                        |
| <b>Note</b>        | Original document. Source document of this translation: D0137978-9/2019-06-04 |            |                                                        |
| <b>Date</b>        | <b>Language</b>                                                               | <b>DCC</b> | <b>Plant/department</b>                                |
| 2019-10-28         | en                                                                            | DA         | WRD Management Support GmbH / Documentation Department |

### Applicable documents

The titles of the documents listed are the titles of the original language versions, with translations of these titles in ( ) where applicable. Document IDs always refer to the original language versions. If the document ID does not contain a revision, the most recent revision of the document applies. This list contains documents concerning optional components if necessary.

| Document ID | Document                                                                                                                                                        |
|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| D0138302    | Datenblatt ENERCON SCADA PDI-OPC (ENERCON SCADA PDI-OPC data sheet)                                                                                             |
| D0140738    | Spezifikation ENERCON SCADA Datenanbindung des ENERCON SCADA Servers (Specification for the ENERCON SCADA system – Data connection of the ENERCON SCADA Server) |

## Table of contents

|            |                                                                             |           |
|------------|-----------------------------------------------------------------------------|-----------|
| <b>1</b>   | <b>Introduction.....</b>                                                    | <b>5</b>  |
| <b>2</b>   | <b>ENERCON SCADA PDI-OPC within the ENERCON SCADA system.....</b>           | <b>6</b>  |
| <b>3</b>   | <b>Functional description .....</b>                                         | <b>7</b>  |
| <b>3.1</b> | <b>Data structure .....</b>                                                 | <b>7</b>  |
| 3.1.1      | Data management.....                                                        | 7         |
| 3.1.2      | Data types .....                                                            | 8         |
| <b>3.2</b> | <b>Data querying by means of Basic Polled Subscription Interaction.....</b> | <b>9</b>  |
| <b>3.3</b> | <b>Namespace in the ENERCON SCADA PDI-OPC .....</b>                         | <b>10</b> |
| 3.3.1      | The Loc (location) node.....                                                | 11        |
| 3.3.2      | The Doc (documentation) node.....                                           | 12        |
| 3.3.3      | The SCADA node.....                                                         | 13        |
| <b>3.4</b> | <b>Transmission of setpoints and control values .....</b>                   | <b>14</b> |
| <b>3.5</b> | <b>OPC session workflow .....</b>                                           | <b>17</b> |
| 3.5.1      | Session start.....                                                          | 17        |
| 3.5.2      | Input of control values .....                                               | 18        |
| 3.5.3      | Data transmission to the ENERCON SCADA system.....                          | 19        |
| <b>3.6</b> | <b>Global reservation of all access for control purposes.....</b>           | <b>21</b> |
| <b>3.7</b> | <b>Processing of the data by the customer .....</b>                         | <b>22</b> |
| <b>3.8</b> | <b>Fail-safety and default values.....</b>                                  | <b>23</b> |
| <b>4</b>   | <b>System requirements .....</b>                                            | <b>25</b> |
| <b>5</b>   | <b>Required maintenance .....</b>                                           | <b>26</b> |
| <b>6</b>   | <b>Scope of supply and services .....</b>                                   | <b>27</b> |
|            | <b>Glossary of terms .....</b>                                              | <b>28</b> |

# 1 Introduction

The ENERCON SCADA PDI-OPC is an OPC-based process-data interface that can be used to evaluate and control individual wind energy converters, wind farms and additional systems online and independently of the ENERCON SCADA Remote software.

The ENERCON SCADA PDI-OPC is an OPC XML-DA server in accordance with OPC Foundation specifications V1.01.

Complete, fault-free transmission of data to the wind energy converters cannot be guaranteed in all cases due to the possibility of communication problems or interference occurring. The start and stop functionality is subject to this restriction.

## **Purpose of this document**

This document describes the use of the ENERCON SCADA PDI-OPC. This document does not describe the basics of the OPC standard. For a detailed description of the data tree and the items used in OPC, please refer to the document D0138302 "Datenblatt ENERCON SCADA PDI-OPC" (ENERCON SCADA PDI-OPC data sheet).

The WEC-related information and data contained in this document are applicable and valid for the following ENERCON wind energy converters:

E-44, E-48, E-53, E-70 E4, E-82 E2, E-82 E4, E-92, E-103 EP2, E-101, E-115, E-115 E2, E-126 EP3, E-115 EP3 E3, E-138 EP3, E-138 EP3 E2, E-126 EP4 and E-141 EP4.

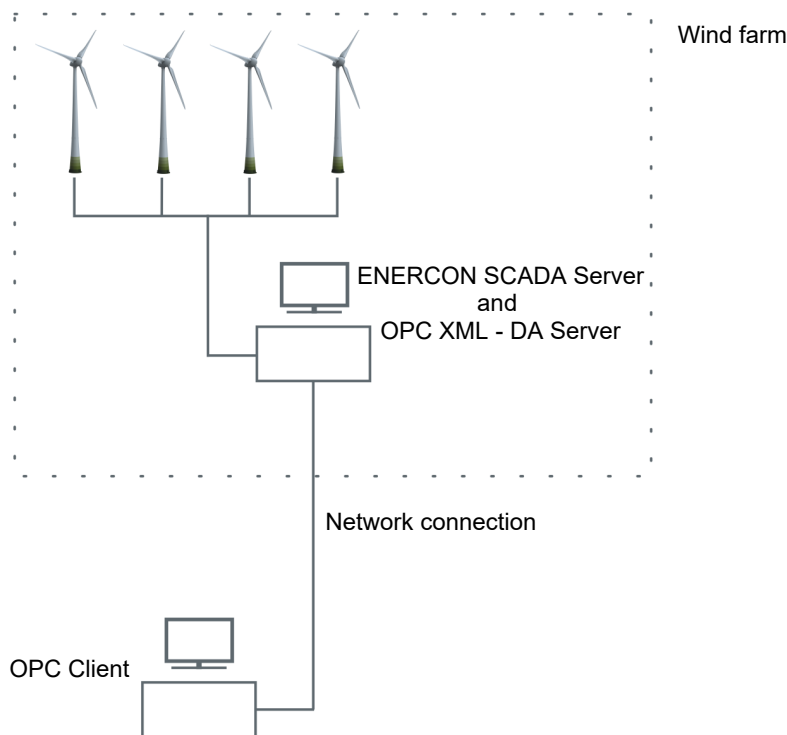
## **User skills**

A basic knowledge of the OPC standard is required. Information about the OPC standard can be found at [www.opcfoundation.org](http://www.opcfoundation.org).

## 2 ENERCON SCADA PDI-OPC within the ENERCON SCADA system

ENERCON SCADA PDI-OPC is implemented as a software solution on the ENERCON SCADA Server and expands its functionality to include that of an OPC XML DA server. ENERCON SCADA PDI-OPC enables online access to wind farm and WEC data, independent of the ENERCON SCADA Remote software. In addition, setpoints can be transmitted.

The OPC XML DA protocol (V1.01) used for that purpose is a non-proprietary, Internet-capable data exchange protocol. Thanks to its simple structure, it is particularly suited for exchanging data located on different networks. Data can also be transmitted over the Internet.



**Fig. 1: ENERCON SCADA PDI-OPC within the ENERCON SCADA system**

The OPC XML DA server supports data buffering for clients (Basic Polled Subscription Interaction; see ch. 3.2, p. 9). Once a client activates this function, the server temporarily stores changes to send them at the client's next request.

Communication failures may cause data inconsistency. To ensure that key data (e.g. status data) can be retrieved from the OPC server in their entirety, the data history is made available for synchronisation purposes. Related data such as status data (status; substatus; information, etc.) are provided as an array, since they are generated at the same point in time and are assigned identical timestamps.

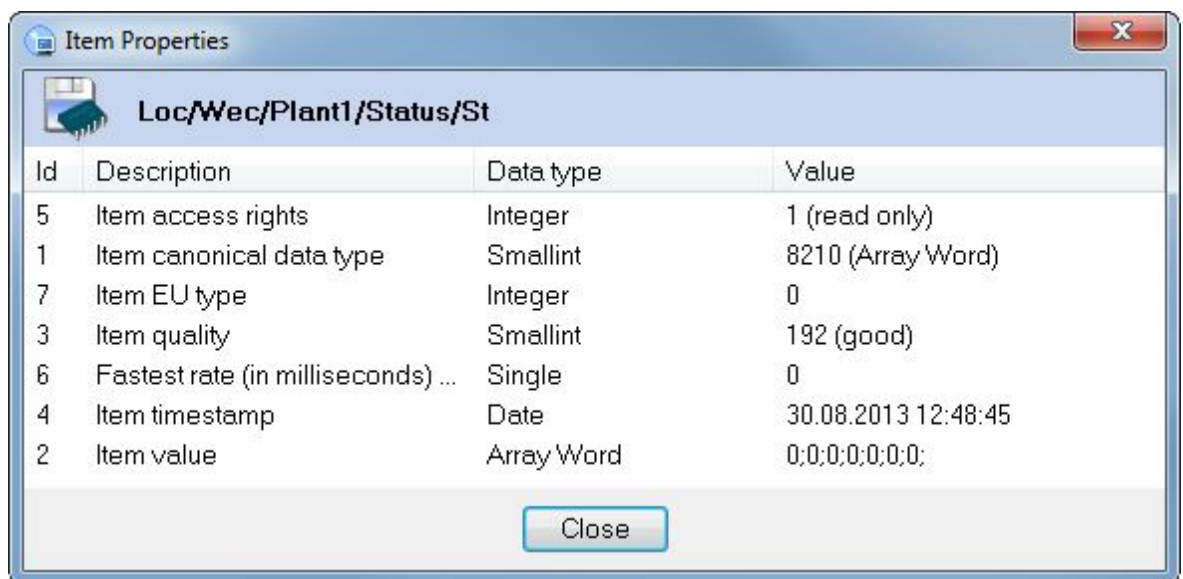
## 3 Functional description

### 3.1 Data structure

#### 3.1.1 Data management

All data sources are managed in the namespace of the OPC server. For the ENERCON SCADA PDI-OPC, the namespace has been set up as a tree structure. Nodes serve as structuring elements of this tree. Items comprise the actual data that the client can access in read-only and/or write mode.

Each item identified in the namespace by a unique name (e.g. 'CalcP') is described by the following value properties:



| Id | Description                        | Data type  | Value               |
|----|------------------------------------|------------|---------------------|
| 5  | Item access rights                 | Integer    | 1 (read only)       |
| 1  | Item canonical data type           | Smallint   | 8210 (Array Word)   |
| 7  | Item EU type                       | Integer    | 0                   |
| 3  | Item quality                       | Smallint   | 192 (good)          |
| 6  | Fastest rate (in milliseconds) ... | Single     | 0                   |
| 4  | Item timestamp                     | Date       | 30.08.2013 12:48:45 |
| 2  | Item value                         | Array Word | 0;0;0;0;0;0;        |

**Fig. 2: Data record of an online value and its item properties**

**Tab. 1: Data record of an online value with value properties**

| ID | Parameter                | Description                                                                                                                         |
|----|--------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| 5  | Item access rights       | Indicates whether the item is read-only or also writeable.                                                                          |
| 1  | Item canonical data type | Indicates the data type of the item.                                                                                                |
| 7  | Item EU type             | Indicates whether the value is an enumerated type (defined states of a group of values).                                            |
| 3  | Item quality             | Indicates the quality of the item (192 (good)/28 (uncertain)).                                                                      |
| 6  | Fastest rate             | Indicates how quickly the value can be updated on the OPC server. If the value is zero, it can change randomly (event-based value). |
| 4  | Item timestamp           | Indicates when the item was written (based on the time zone (UTC) configured on the server).                                        |
| 2  | Item Value               | Indicates the value of the item in the form of the data type.                                                                       |

The *Fastest rate* parameter applies only to online data. Since report, status, availability and control data are event-based, the fastest rate is set to zero for these items. If the WEC is inactive, or in the event of communication faults, the value for item quality is set to *uncertain*. For report data, the respective value will be set to *uncertain* if data points are not populated with data.

### 3.1.2 Data types

The OPC server provides the following data types:

#### Online values

Online values are instantaneous values that the OPC server refreshes as often as permitted by the wind farm's ENERCON SCADA System. Their informative value is limited so that there is no need to store them on the ENERCON SCADA Server.

The data refresh rate depends on the number of wind energy converters installed in the wind farm, the structure of the wind farm data bus, and – primarily – the communication hardware selected. The structure of each specific project defines the highest possible transfer rate. It is communicated by the system and can be viewed on the client. The shortest refresh interval within ENERCON SCADA PDI-OPC is one second.

Fixed parameters such as the serial number of a specific wind energy converter also belong to the online values group.

#### Report data

Report data are generated through various measurements (data loggers) as part of the WEC monitoring process. These include WEC status data, as well as mean values for a specific time period. The latter include data collected on the ENERCON SCADA Server.

Mean values are commonly computed for one minute, ten minutes, one day, one week and one month. All of these are based on one-minute mean values. All 10-minute means (and longer intervals) are stored on the ENERCON SCADA Server. The system retains historical report data. The OPC server provides not only current values but also values from past time intervals. This ensures the consistency of the data on the ENERCON SCADA Server and the customer's PC, even if online communication is defective.

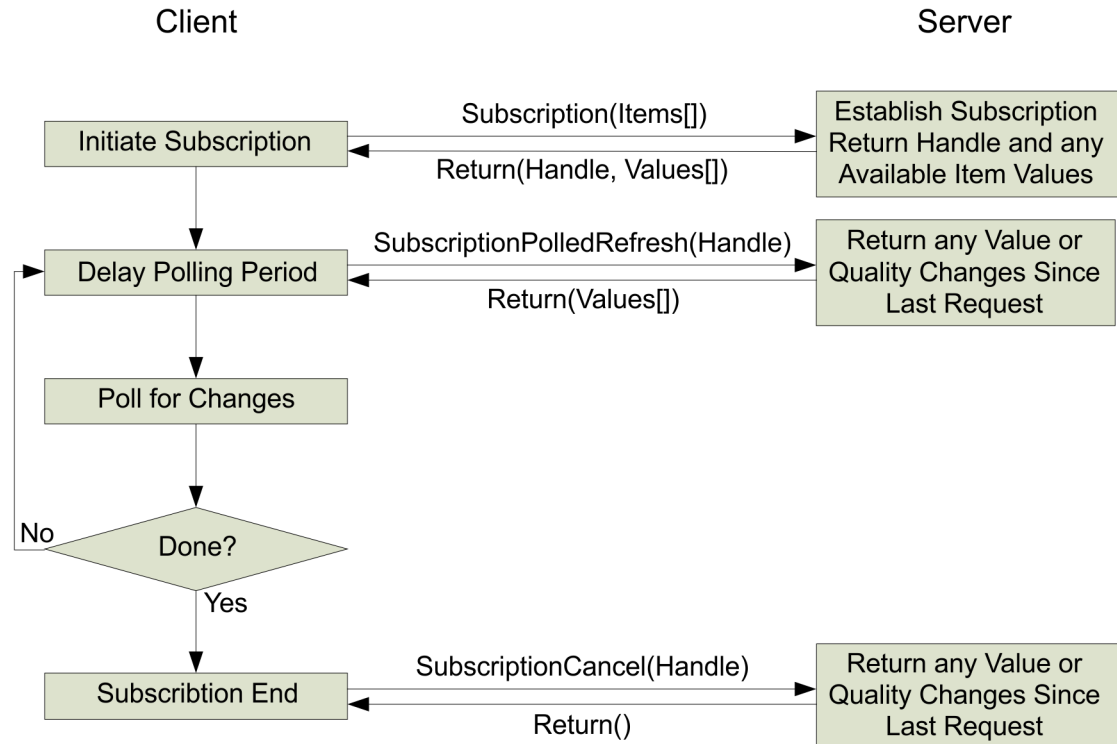
If the ENERCON SCADA Server cannot poll any data from the wind energy converters and devices, it will not be able to compute any 10-minute means. Therefore, no report data will be available in OPC for that period.

#### Control data

Control data can be used by the client to change wind farm or wind energy converter parameters. This includes open-loop control of the wind farm control system as well as starting and stopping individual WECs.

### 3.2 Data querying by means of Basic Polled Subscription Interaction

In accordance with the OPC XML-DA V1.01 specification, data are queried using 'basic polled subscription interaction'. This is a method for exchanging data between server and client software. The diagram below illustrates the principle of this method:



**Fig. 3: Basic Polled Subscription diagram**

The client software initialises a subscription with the data points to be requested on the server, which are polled cyclically by the client. During the 'delay polling period', the server records the changes to the requested data points in order to make them available to the client software during the polling refresh. Using this form of data compression, only the changed data points are communicated to the client. In order to terminate a subscription, the client must notify the server. Each installed subscription reserves resources on the server in the form of a dedicated thread.



A maximum of 100 subscriptions is permitted so as not to compromise the operational readiness of the ENERCON SCADA PDI-OPC. The server will close any subscription whose data have not been collected after the polling period has expired (Subscription Ping Rate).

### 3.3 Namespace in the ENERCON SCADA PDI-OPC

The namespace describes the structure of the data points in the ENERCON SCADA PDI-OPC. Whereas the structure of the transmitted data is specified by ENERCON, their graphical representation on the client depends on the software used by the customer for that purpose. Software programs that represent the nodes as file folders forming a tree structure are commonly used. Items in these folders are shown as files.

The screenshots used in this document were made using Kassl OPC Explorer 1.36.

There are three nodes located in the root folder of the ENERCON SCADA PDI-OPC:

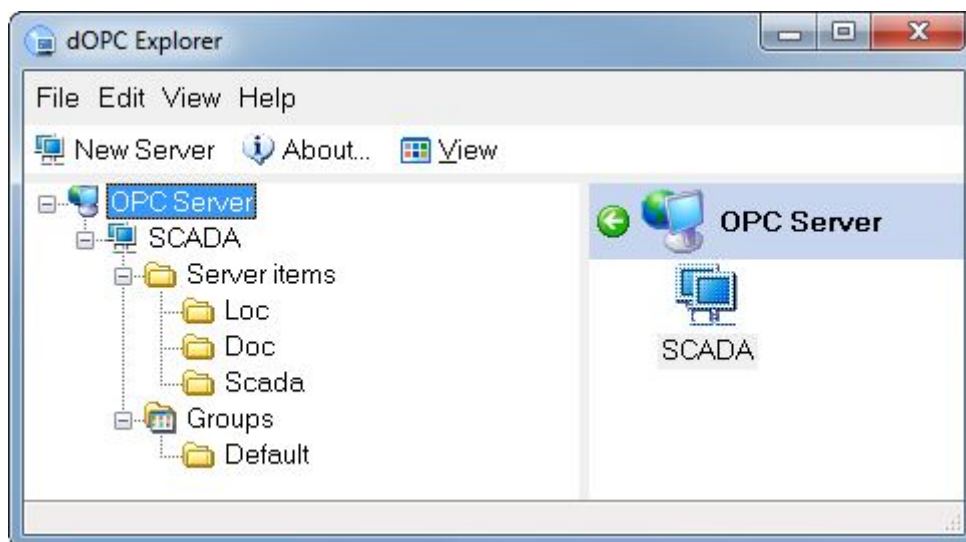
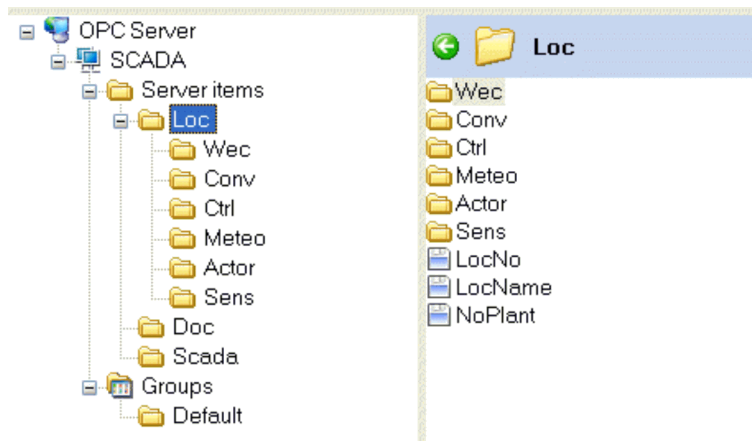


Fig. 4: Nodes in the root folder of the ENERCON SCADA PDI-OPC namespace

Tab. 2: Root-node nodes

| Node  | Description                                                                                                                                                          |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Loc   | All plant data for a wind farm (e.g. data from wind energy converters; open-loop and closed-loop control systems such as RTU or FCU; METEO weather-measurement data) |
| Doc   | Short description of the items in ENERCON SCADA PDI-OPC. A short description of the items is provided based on the data structure of the Loc folder.                 |
| Scada | SCADA-specific items such as time                                                                                                                                    |

### 3.3.1 The Loc (location) node



**Fig. 5: The Loc (location) node**

The term *location* refers to a wind farm. All information referring to this location is available under the *Loc* node. Here, data sources include not only the wind energy converters in the wind farm, but also all other components of the ENERCON SCADA system, such as ENERCON SCADA METEO, ENERCON SCADA RTU and similar systems.

A detailed description of the nodes and items in the *Loc* node can be found in document D0138302 "Datenblatt ENERCON SCADA PDI-OPC" (ENERCON SCADA PDI-OPC data sheet).

**Tab. 3: The Loc (location) node**

| Node/item | Description                                                                                                            |
|-----------|------------------------------------------------------------------------------------------------------------------------|
| Wec       | Data from the wind energy converters                                                                                   |
| Sens      | Data from the sensor systems (e.g. medium-voltage data acquisition)                                                    |
| Conv      | Data from the converter systems (e.g. solar converter)                                                                 |
| Ctrl      | Data from the open-loop control systems (e.g. ENERCON SCADA RTU)                                                       |
| Actor     | Data from the actors (e.g. ENERCON SCADA PDI)                                                                          |
| Meteo     | Data from the ENERCON METEO weather recording system                                                                   |
| LocNo     | Wind farm number assigned by ENERCON; identical to the ID used for identification in the ENERCON SCADA Remote software |
| LocName   | Unique name of the wind farm defined by the customer                                                                   |
| NoPlant   | Number of all plants in this wind farm, including all devices connected to the ENERCON SCADA system                    |

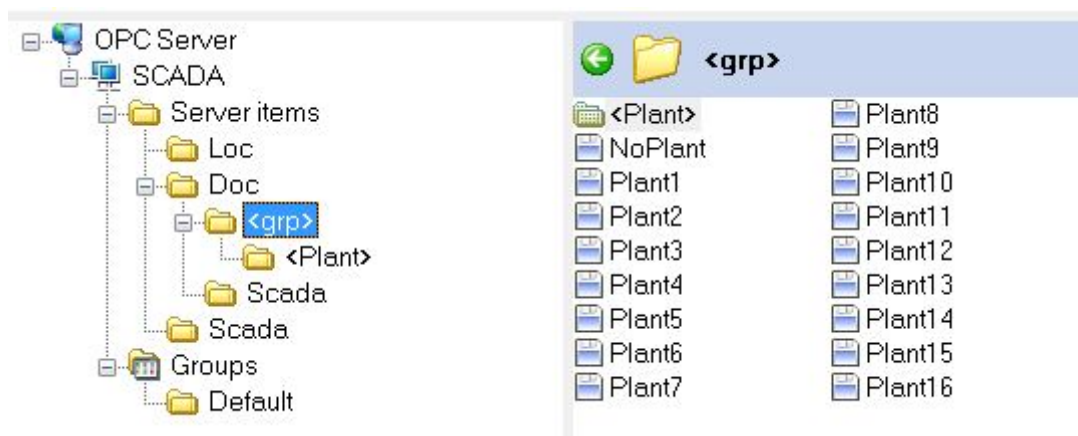
### 3.3.2 The Doc (documentation) node

| Id | Description                   | Data type | Value                  |
|----|-------------------------------|-----------|------------------------|
| 5  | Item access rights            | Integer   | 1 (read only)          |
| 1  | Item canonical data type      | Smallint  | 8 (OleStr)             |
| 7  | Item EU type                  | Integer   | 0                      |
| 3  | Item quality                  | Smallint  | 192 (good)             |
| 6  | Fastest rate (in milliseco... | Single    | 0                      |
| 4  | Item timestamp                | Date      | 30.08.2013 12:48:54    |
| 2  | Item value                    | OleStr    | No. of historical data |

**Fig. 6: Explanation of the corresponding *NoSt* item in the *Loc* node**

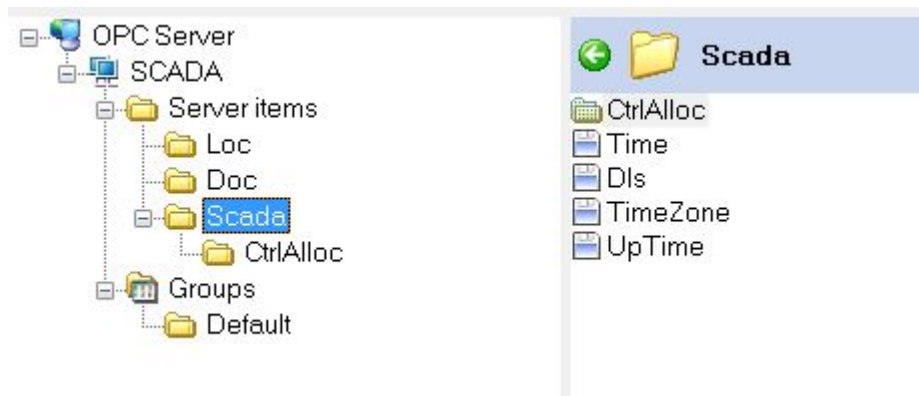
The items in the *Doc* node provide explanations of the corresponding items in the *Loc* node. This can also be used to look up abbreviations used for nodes and items in the ENERCON SCADA PDI-OPC.

The structure of the *Doc* node is essentially identical to that of the *Loc* node. The *<grp>* (group) node comprises all wind energy converters and ENERCON SCADA system devices for one wind farm. The total number is again reflected in the *NoPlant* item.



**Fig. 7: The *Doc* (documentation) node**

### 3.3.3 The SCADA node



**Fig. 8: The Scada node**

The *Scada* node contains SCADA-specific items resulting from the project-specific functions of the ENERCON SCADA system.

**Tab. 4: The Scada node**

| Node/item           | Description                                                                                                                                                                                                                                                                                                                                       |
|---------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CtrlAlloc (example) | Items for global allocation of all access for control purposes                                                                                                                                                                                                                                                                                    |
| Time                | Depending on the configuration, the following times are transmitted: <ul style="list-style-type: none"> <li>■ UTC activated<br/>The value transmitted is the coordinated universal time (UTC; format dd.mm.yyyy hh.mm.ss).</li> <li>■ UTC deactivated<br/>The value transmitted is the local server time (format dd.mm.yyyy hh.mm.ss).</li> </ul> |
| DIs                 | Flag for daylight saving time                                                                                                                                                                                                                                                                                                                     |
| TimeZone            | Time zone east of the prime meridian in minutes (German time zone = +60)                                                                                                                                                                                                                                                                          |
| UpTime              | Start-up time of the OPC server, e.g. after a grid fault. This point in time is not identical to the time of installation.                                                                                                                                                                                                                        |

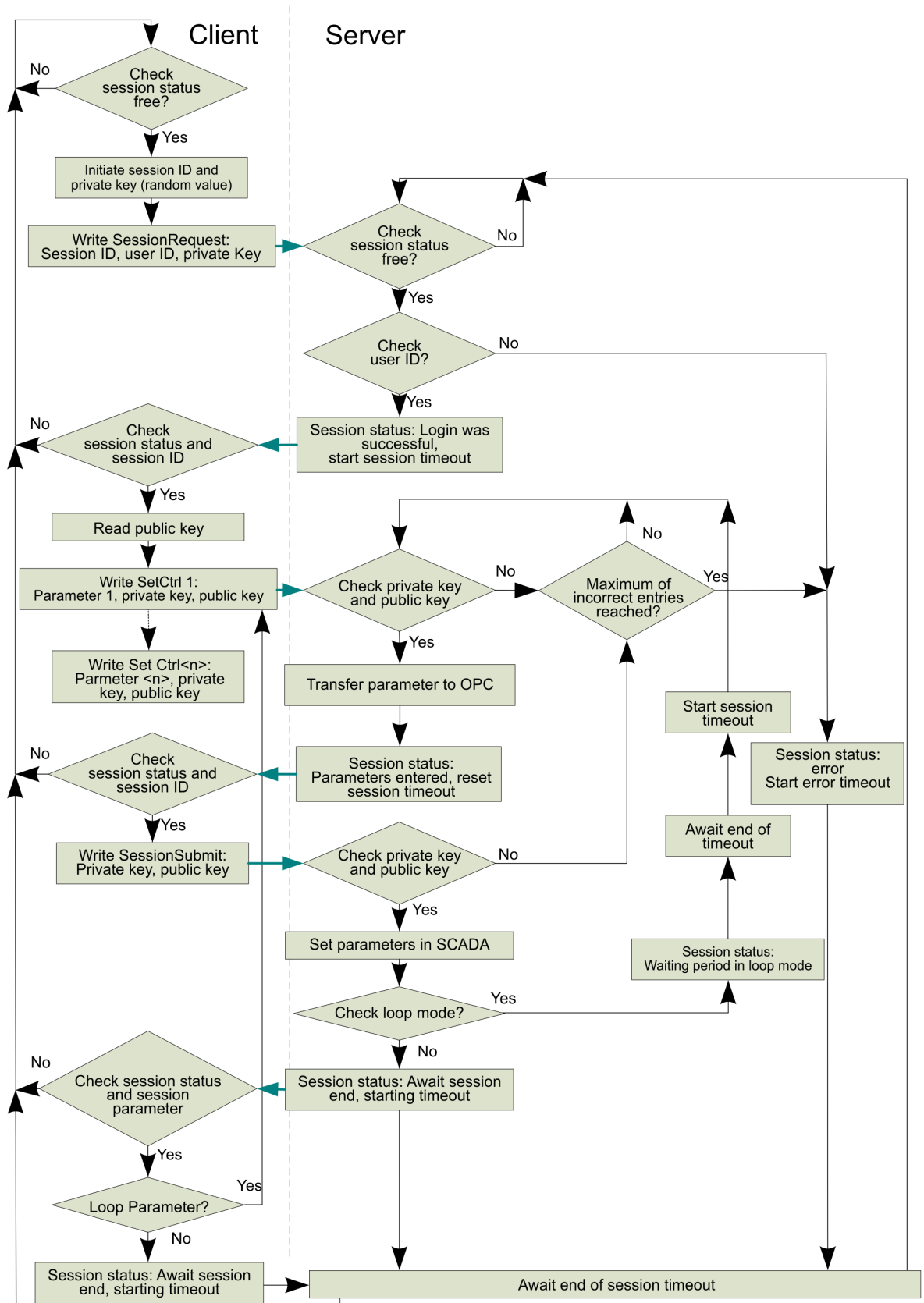
### 3.4 Transmission of setpoints and control values

The ENERCON SCADA PDI-OPC supports the transmission of setpoints and control values, such as for starting and stopping a wind energy converter or for providing the wind farm controller (e.g. ENERCON SCADA RTU and ENERCON SCADA FCU) with setpoints.

ENERCON SCADA PDI-OPC processes these setpoints and control values in sessions. This ensures that actions are identifiable and that only one client at a time can perform one action.

This mechanism is exclusively intended to manage and identify access by communication partners. It is not a security mechanism like VPN or SSL encryption.

The diagram below presents the session workflows in detail:



**Fig. 9: OPC session diagram**

In simple terms, four process steps for transmitting parameters or control values to the OPC server are run:

- A session is reserved for access by specifying a session ID, a user ID and a private key. The respective values are written to an array.
- The appropriate parameters are entered into the items. They do not take effect yet.
- Parameters become active once the system has acknowledged their transfer.
- The timeout must have elapsed before a new session can be reserved.

There are two types of session:

### Single Session

As described above, a single session comprises four steps. It is typically used to set parameters or control values.

### Loop session

A loop session is used to receive cyclical data from clients. This forces the continuous communication with the clients and requests 'alive' signals from the clients. At first, this session progresses just like a single session. However, once the parameters have been entered into the system ('session submit'), the session will return to parameter input without making another reservation. Typically, one 'session submit' is enough to execute the 'alive' signal.

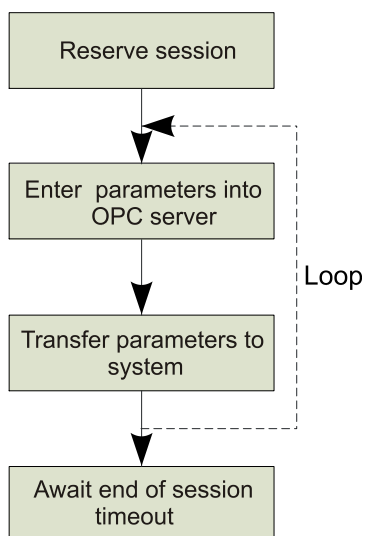


Fig. 10: Loop session diagram

### Permissions

In order for different parties to be able to transmit setpoints externally, the relevant permissions must be defined for each individual project and integrated into the ENERCON SCADA system. The definition of the permissions must be agreed in advance with ENERCON Service.

For security reasons, authentication of the sender's identity is required for each setpoint transmission.

### Setpoints

ENERCON offers setpoints for the following functions:

- Switching a wind energy converter on or off (Start/Stop)
- Wind farm controllers

## Timeouts

The duration of a session is defined by the regular timeout. Each time a setpoint or parameter has been modified, the extended timeout starts: this prevents another change for a specific time period. This dynamic interval ensures that electrical values have been fully adjusted before the next setpoint modification takes place. In the event of an incorrect input, another input will be prevented for a specified time period.

**Tab. 5: Overview of timeouts**

| Timeout                                                     | Value                 |
|-------------------------------------------------------------|-----------------------|
| Normal timeout during a session                             | Can be set in minutes |
| Extended timeout after setting a parameter/control value    | Can be set in minutes |
| Session end timeout (delay for new reservation)             | 10 s                  |
| Session end timeout (delay for new reservation after stop)  | 360 s                 |
| Session end timeout (delay for new reservation after start) | 0 s                   |
| Error timeout (after three incorrect entries)               | 300 s                 |
| User ID error timeout                                       | 300 s                 |

## 3.5 OPC session workflow

The following section explains the workflow of an OPC session, using the transmission of a setpoint for the function *Start/Stop* of a wind energy converter.

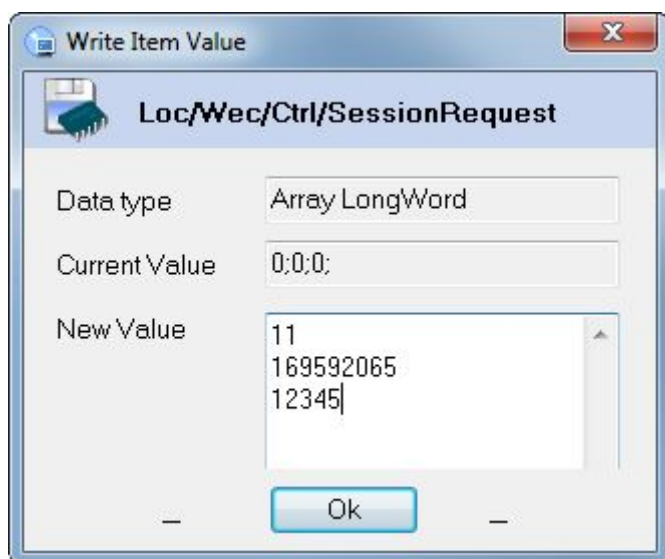
### 3.5.1 Session start

In order to start a session, the *SessionRequest* item must be written to the OPC server. This item can be found in the *Ctrl* node of the respective wind energy converter.

The following values must be written to the array in the sequence defined below:

**Tab. 6: Description of values for one session**

| Value       | Description                                                                                                                                                                                                                         |
|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Session ID  | Random value from the client, which the client can use to determine whether a reservation was successful. In the example shown in the illustration below, the value of the session ID is 11. This value can be read by the clients. |
| User ID     | The user ID represents a user in the ENERCON SCADA system. SCADA users and their permissions are created in the ENERCON SCADA system by ENERCON Service IT. This value cannot be read and is therefore invisible to other clients.  |
| Private key | The private key is a random value and a subkey for access to the respective session. This value cannot be read and is therefore invisible to other clients.                                                                         |



**Fig. 11: Writing the *SessionRequest* item**

Once the values for the session ID, user ID and private key have been written to the *SessionRequest* item, the session state that can be viewed in the *SessionState* item changes to 1 (session reserved) if the login was successful. Otherwise, an error code appears, and a timer starts in order to prevent any other login attempt for a specific period. This time period is defined by ENERCON.

**Tab. 7: Feedback code**

| Feedback code | Description                          |
|---------------|--------------------------------------|
| 0             | Session free                         |
| 1             | Session reserved                     |
| 2             | Parameter input                      |
| 3             | Wait time in loop mode               |
| 4             | Session-end wait time                |
| 5             | Session blocked (global reservation) |
| 108           | Busy                                 |
| 109           | Access denied                        |
| 121           | Illegal value                        |
| 174           | Incorrect user ID                    |
| 175           | Insufficient access rights           |

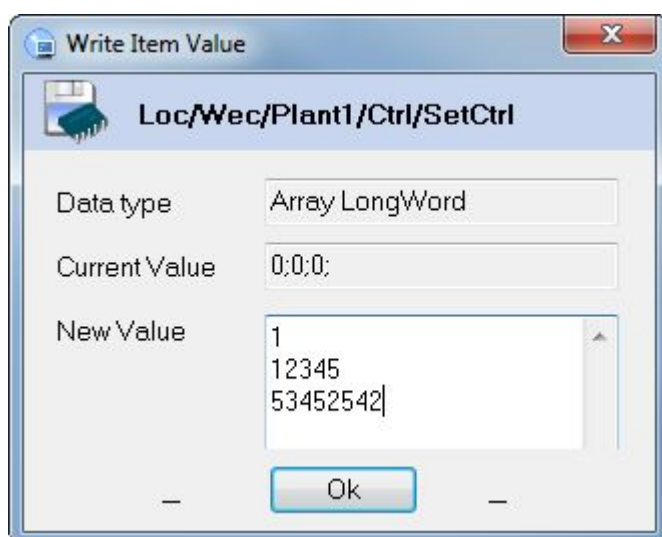
If the login was successful, a session timeout is initiated, and a public key for the session is written to the *SessionPubKey* item.

All operations carried out during this session use the private key and the public key. To confirm that the user has actually reserved this session, the user must verify their session ID.

### 3.5.2 Input of control values

The control values for a wind energy converter are defined in the *SetCtrl* item. The following values must be written in this defined sequence:

|               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Control Value | <p>One of the following control values must be used for starting or stopping the wind energy converter.</p> <p>0 = WEC start</p> <p>1 = WEC stop at 60° blade angle</p> <p>2 = WEC stop at 90° blade angle</p> <p>3 = WEC gradient stop at 60° blade angle</p> <p>4 = WEC gradient stop at 90° blade angle</p> <p>5 = Ice detection stop</p> <p>6 = Shadow casting stop</p> <p>7 = Animal protection stop at 60° blade angle</p> <p>8 = Animal protection stop at 90° blade angle</p> |
| Private Key   | The private key selected by the user at the start of the session                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Public Key    | The public key made available by the server at the start of the session                                                                                                                                                                                                                                                                                                                                                                                                               |



**Fig. 12: Writing the SetCtrl item**

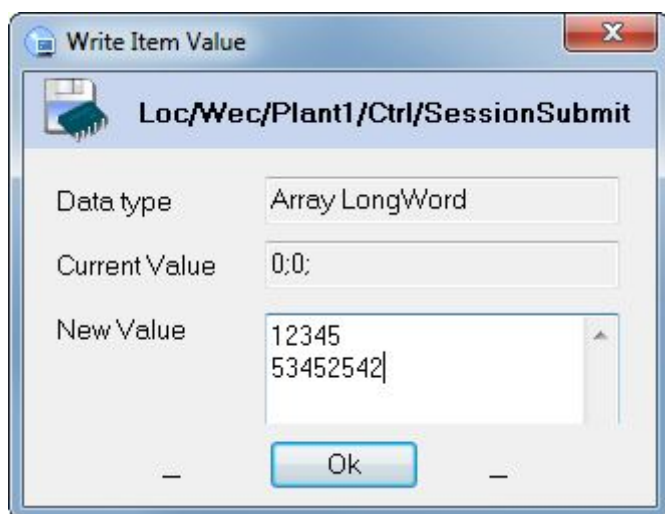
The private key and the public key must match the keys used at the start of the session. If the user enters one of these keys incorrectly three times, the corresponding error code will be displayed in the *SessionState* item, and a timeout will be started. This timeout prevents the input of new data for a specific time period.

Once the data have been entered correctly, the *SessionState* changes from *SessionReserved* to *SessionParameterEntered*. The timeout for that session will also be reset so that the user will still be able to edit the values.

### 3.5.3 Data transmission to the ENERCON SCADA system

After the values have been entered into the OPC client, they must be transmitted to the OPC server in the ENERCON SCADA system for execution. This process is started by writing the private key and the public key to the *SessionSubmit* item.

The private key and the public key must match the keys used at the start of the session. If the user enters one of these keys incorrectly three times, the corresponding error code will be displayed in the *SessionState* item, and a timeout will be started. This timeout prevents the input of new data for a specific time period.



**Fig. 13: Writing the *SessionSubmit* item**

After the private key and the public key have been transmitted, the session status changes to *WaitSessionEnd* (4). After the status change, a timer is started in order to prevent the user from entering more data. When the wind energy converter has accepted the modified control data, the *Ctrl* item displays the status of the WEC to the user.

**Tab. 8: Feedback code**

| Feedback code | Description                                              |
|---------------|----------------------------------------------------------|
| 0             | WEC start                                                |
| 1             | WEC stop at 60°                                          |
| 2             | WEC stop at 90°                                          |
| 121           | Wrong value                                              |
| 129           | WEC stopped at 60° (with higher priority). Cannot start. |
| 130           | WEC stopped at 90° (with higher priority). Cannot start. |
| 255           | Communication error. A timeout has occurred.             |

### 3.6 Global reservation of all access for control purposes

The 'global reservation' function can be used to reserve all accesses for control purposes by means of the ENERCON SCADA PDI-OPC for a single user (user ID). Following initial reservation of the session (*SessionRequest*), there is merely a cyclical run of the session (*SessionSubmit*) in a loop session.

The reservation is associated with a priority assigned to each ENERCON SCADA PDI-OPC user (each user ID) in the ENERCON SCADA system. Level 0 has the highest priority.

Users with higher priority levels can cancel the existing reservations of users with lower priority levels and can thus take control of the access for control purposes in the ENERCON SCADA PDI-OPC.

Following successful reservation, all *SessionState* items display this reservation with an indication of the session ID and the public key. Access for control purposes by means of these private and public keys is now possible without having to make the respective reservation in advance.

Data are transmitted in a loop session. The *UserName* item (login name of the user in the ENERCON SCADA system) resides in the *Scada/CtrlAlloc* node.

### 3.7 Processing of the data by the customer

The transmission of the data provided by the ENERCON SCADA Server (= OPC XML-DA server) in the wind farm to the customer's IT system is the responsibility of the customer. ENERCON Service IT offers support in selecting the appropriate data link.

The customer's IT system must include a software that is designed for data exchange using the OPC XML-DA protocol. A wide range of commercial software applications is available on the market. Customers can decide individually which data to select, display and store on their systems.

The OPC XML-DA server in the wind farm is designed so that customers can save exactly the same data to their local systems that exists in the wind farm, even if online communication is defective. This is achieved by providing the entire history in addition to current online and mean values. In this way, if data transmission is interrupted, customers can fill in any gaps in their data at a later stage. However, this process must be initiated by the OPC client (=customer) because the server cannot determine which gaps might have occurred in the customer's data.

As a significant portion of the communication and data processing takes place outside the wind farm, and thus beyond ENERCON's responsibility, customers have no legal claim to complete data transmission.

### 3.8 Fail-safety and default values

If setpoint specification via the ENERCON SCADA PDI-OPC interface of the ENERCON SCADA Server is disrupted, the ENERCON SCADA Server cannot receive any new setpoints via the interface. Setpoint specification can be disrupted by the following events, for example:

- Interface errors
- Disrupted communication with the unit specifying the setpoint
- Server is offline

#### Interface malfunction

If communication with the interface or the interface itself is faulty and the server is online, the controller used (ENERCON SCADA FCU or RTU) will continue closed-loop control using the last setpoints received.

#### Server offline - behaviour of FCU

If control is via the ENERCON SCADA FCU and the ENERCON SCADA Server is shut down due to a fault for instance (server offline), the FCU will continue operation using the last setpoints received.

#### Server offline – behaviour of the FCU E2

If closed-loop control is realised via the ENERCON SCADA FCU E2 and the ENERCON SCADA Server is shut down due to a fault (server offline), for instance, the FCU E2 will continue operation using the last setpoints received or will use saved default setpoints.

#### Server offline – behaviour of the RTU

If closed-loop control is realised via the ENERCON SCADA RTU and the ENERCON SCADA Server is shut down due to a fault (server offline), for instance, control via the RTU will not be possible for as long as the server is down.

If parameter 1308 for the wind energy converters is deactivated, the wind energy converters will use the control values stored in parameters 1209, 1210 and 1400. The control values will not be stored by the ENERCON SCADA Server if parameter 1308 is deactivated. The control values stored will then be the configured default values.

If parameter 1308 *SCADA farm control* (parameter for wind energy converters) is activated, the ENERCON SCADA Server will save the last valid control value to the following parameters of the wind energy converters:

- 1209 *dyn. react.pow.(Q/P): (farm contr. faulty)*
- 1210 *static react. pow.: (farm contr. faulty)*
- 1400 *maximum power (farm contr. faulty)*

To ensure that any slight changes to RTU control values do not result in permanent changes to the last valid control values stored in the wind energy converters, the control values are not written until a specified deadband has been exceeded/undershot. The following values apply here:

- P:  $\pm 10.01 \%$
- Q:  $\pm 10.01 \%$
- Q/P:  $\pm 0.170 (9.65^\circ)$

### **Checking the setpoint**

An evaluation of whether a setpoint has been correctly received and processed by the FCU, FCU E2 or RTU can be made via the feedback data points for the FCU/FCU E2/RTU interfaces for the relevant setpoint.

Checking the correct receipt and adoption of a setpoint is the task of the entity that specified the setpoint (e.g. the operator/owner).

## 4 System requirements

To ensure optimum communication between the ENERCON SCADA Server and the outside world via OPC XML-DA, setting up a permanent online connection between the server and the client is recommended. The specific implementation is at the customer's discretion.

Since the frequency with which online values on the OPC server are updated depends on the speed of the wind-farm data bus, the fastest possible connection between the individual wind energy converters, the ENERCON SCADA Server and other, optional, components should be used here.

More detailed information on the preconditions for the data connection of the ENERCON SCADA Server can be found in document D0140738 "Spezifikation ENERCON SCADA Datenanbindung des ENERCON SCADA Servers" (Specification for the ENERCON SCADA system – Data connection of the ENERCON SCADA Server).

The customer must grant ENERCON Service access via the following port:

**Tab. 9: Port to be enabled**

| Port | Direction | Purpose               |
|------|-----------|-----------------------|
| 6010 | Incoming  | ENERCON SCADA PDI-OPC |



For security reasons, only private IP addresses that have been configured for OPC access in the SCADA system will be unblocked in the internal SCADA firewall. Only the static IP addresses configured in this manner can access the OPC server. Public IP addresses cannot access or be enabled in the internal SCADA firewall.

### Software compatibility

ENERCON SCADA version 2.0 or greater (Linux-based operating system)

The ENERCON SCADA Server must support the Linux operating system and must have the performance features required by the ENERCON SCADA software. Especially with existing wind farms, the installed ENERCON SCADA Server may need to be replaced. Any existing wind farm data can be copied to the new server.

### Safety

ENERCON SCADA PDI-OPC may only be operated if all communication links are secured by a VPN tunnel.

## 5 Required maintenance

As long as the ENERCON SCADA PDI-OPC is operating correctly, no maintenance is required. The hardware of the ENERCON SCADA Server is subjected to a visual inspection that forms part of the maintenance on the ENERCON SCADA system, provided a maintenance contract or ENERCON PartnerKonzept contract with ENERCON is in place. The ENERCON SCADA PDI-OPC is not part of the ENERCON PartnerKonzept.

Devices installed between the ENERCON SCADA PDI-OPC interface of the ENERCON SCADA Server and the customer's client device are not included in the maintenance provided by ENERCON Service.

## **6 Scope of supply and services**

The ENERCON SCADA PDI-OPC is an optional component and is not included in the regular scope of supply of an ENERCON wind farm.

If agreed, the associated hardware and software are supplied in the form of the ENERCON SCADA Server. Externally, the presence of the ENERCON SCADA PDI-OPC is visible only in the form of the RJ45 network socket.

A detailed description of the nodes and items can be found in document D0138302 "Datenblatt ENERCON SCADA PDI-OPC" (ENERCON SCADA PDI-OPC data sheet).

## Glossary of terms

|                            |                                                                                                                                                                                                                                                                               |
|----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Array</b>               | In computer science, an array is a data structure. Using a field, data of a typically standardised type can be stored in computer memory in an arranged manner. Data access is made possible by using an index.                                                               |
| <b>Client</b>              | Computer program on a terminal device or a terminal device that connects to a server in order to make use of services running on the server. Various services can be accessed, depending on the type of client.                                                               |
| <b>ENERCON SCADA METEO</b> | The ENERCON SCADA METEO system serves to acquire and analyse meteorological data in combination with ENERCON SCADA.                                                                                                                                                           |
| <b>ENERCON SCADA RTU-C</b> | Wind farm controller that can control a wind farm relative to the actual parameter at the NCP.                                                                                                                                                                                |
| <b>Flag</b>                | In computer science, a flag is a binary variable or a status indicator. A flag can be used to identify specific states.                                                                                                                                                       |
| <b>Item</b>                | Within the context of OPC, the combination of a (physical) value and its properties is called an Item.                                                                                                                                                                        |
| <b>Namespace</b>           | A namespace is an abstract container providing context for the items (names, or technical terms, or words) it holds. The use of namespaces allows disambiguation of homonym items having the same name (for instance, identical telephone numbers with different area codes). |
| <b>Node</b>                | Within the context of OPC, a folder/directory for structuring the OPC namespace file system. A node can contain both items and additional nodes.                                                                                                                              |
| <b>OPC XML-DA</b>          | Interface standard of the OPC Foundation which is used for XML-based exchange of data between computer systems, particularly via the Internet, among other things.                                                                                                            |
| <b>Protocol</b>            | Definition of standards and conventions for trouble-free data transmission between participants in communication.                                                                                                                                                             |
| <b>Server</b>              | A server is a program that communicates with other programs and can provide special services and data to the latter.                                                                                                                                                          |
| <b>Thread</b>              | In computer science, a thread of execution or a processing sequence during execution of a program. Threads enable parallel processing.                                                                                                                                        |